

Standard Form (Canonical Form)

Video Lesson

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Lesson Transcript

In this lecture, we will look at standard forms in first-order logic or otherwise known as canonical forms.

Standard Forms of Predicate Calculus



Need this for automated theorem proving:

- 1 Prenex Normal Form: arranged all quantifiers at the front of the formula : use De Morgan's rules (p. 193)
- 2 Convert the non-quantifier part (called the **matrix**) into Conjunctive Normal Form
- 3 Skolemization: eliminate existential quantifiers by introducing **Skolem constants** or **Skolem functions**.

Result:

$\forall x_1 \forall x_2 \dots \forall x_n (\text{CNF})$

In propositional logic, we only had to turn the formulas into conjunctive normal form. However, in first-order logic, we have to follow several steps.

These steps are needed for automated theorem proving.

Those three steps are prenex normal form, conjunctive normal form, and skolemization. For the first step, prenex normal form, we must arrange all quantifiers to the front of the formula. We will use De Morgan's rule and other laws to achieve this. Once we have all the quantifiers in the front, then we need to turn the remaining part into conjunctive normal form.

So, this part is identical to the propositional logic case.

Finally, through the process of skolemization, we can eliminate all existential quantifiers.

As a result, the standard form will have this form. It will begin with all the universally quantified variables and the universal quantifiers, followed by the formula written in conjunctive normal form.

Prenex Normal Form

A formula $\mathcal{A}$ in first-order logic is in **Prenex Normal Form** iff the formula is in the form:

$\underbrace{Q_1 x_1 \ Q_2 x_2 \ \dots \ Q_n x_n}_{\text{Prefix}} \underbrace{(M)}_{\text{Matrix}}$ where Q_i is $\forall$ or $\exists$, and M contains no quantifiers.

A formula F in first-order logic is in Prenex Normal Form if and only if the formula is in this form, $Q_1 x_1 Q_2 \dots Q_n x_n M$, where Q_i is either the universal quantifier symbol $\forall$ or the existential quantifier symbol $\exists$.

x_i , in this case, is the variable associated with that particular quantifier, and M contains the rest of the formula. These variables, x_1, x_2 to x_n , would appear in some of these predicates and functions included in this matrix.

Conjunctive Normal Form

Analogous to propositional logic.

- a list of clauses (disjunction of literals)
- r -literal clause, unit clause, and empty clause.

The next step is the Conjunctive Normal Form, which is analogous to the propositional logic case. So, we have a list of clauses, each clause being a disjunction of literals. There could be r -literal clauses, unit clauses with a single literal, or empty clauses without any literal.

Quantifier Equivalences: Converting to Prenex Normal Form

Equivalence formulas ($Q = \forall$ or $\exists$):

- $(Qx F(x)) \vee G = Qx (F(x) \vee G)$
 $(Qx F(x)) \wedge G = Qx (F(x) \wedge G)$
- $\neg(\forall x F(x)) = \exists x (\neg F(x))$
 $\neg(\exists x F(x)) = \forall x (\neg F(x))$
- $(\forall x F(x)) \wedge (\forall x G(x)) = \forall x (F(x) \wedge G(x))$ $(\exists x F(x)) \vee (\exists x G(x)) = \exists x (F(x) \vee G(x))$
- $(Q_1 x F(x)) \vee (Q_2 x H(x)) = Q_1 x Q_2 z (F(x) \vee H(z))$ $(Q_1 x F(x)) \wedge (Q_2 x H(x)) = Q_1 x Q_2 z (F(x) \wedge H(z))$

To convert a formula into Prenex Normal Form, we need to learn about quantifier equivalences.

To look at these examples, let us first assume that this Q symbol can be any of these two quantifier symbols, either the universal or the existential quantifier. So, $Q_x F(x)$ or G , where G

does not contain this variable x . Then we can basically move these parentheses in to include $F(x)$ or G so that this quantifier Q with associated variable x would have a scope across the entire formula.

Same for conjunction. If this formula G does not contain the variable x , then again, we can extend the scope of this quantifier with the variable x across the entire expression.

The next rule is very important because we want to move all of the negations inside of these quantifiers. If we have $\neg \forall x F(x)$, then we can turn it into there exists an x such that $F(x)$ is not true. We can think of this negation coming inside the quantifier symbol, and as we do that, this quantifier is flipped to the opposite quantifier.

The same is true for the existential quantifier. So, when we have a negation across this existentially quantified expression, then we can bring in this negation inside and flip this quantifier to be universal.

Next, when these universally quantified variables are repeated, the scope for the first one is across $F(x)$, and then the scope for the second one is, of course, $G(x)$. But when there is a conjunction in the middle, we can collapse this into a single expression—using only x as the universally quantified variable.

So, $\forall x F(x) \wedge \forall x G(x) = \forall x G(x) \wedge F(x) \wedge G(x)$.

So, this does not work when the connective in the middle is a disjunction.

In the second case, we have existentially quantified variable x over $F(x)$, and then the same variable is repeated, but it has the scope only over $G(x)$. Then in the case where this connective in the middle is a disjunction, we can collapse that to a single existentially quantified variable x , having scope over both $F(x)$ and $G(x)$ with a connective disjunction in the middle.

In general, regardless of the quantifier, if the variable is repeated with the different scope shown here, we can combine these and pull all of these together to the front as long as we rename the redundant variable name. So, instead of having x and x , if we rename the second x to be z so that the second part becomes quantifier two $z H(z)$, then we can pull out this second quantifier to the front and put this formula from the second part by replacing x with z so that we have Quantifier 1 x , Quantifier 2 z over $F(x)$ or $H(z)$.

So, it is important to note that this variable name has been changed, and this quantifier survives with that same changed variable name. So, contrasted to the above, here we just use the single universal quantifier variable x and a single existentially quantified variable x to cover the entire scope.

So, the same is true here for disjunction and conjunction. So again, if we have to repeat the

variable x , then we can rename the second variable x to z or any arbitrary variable that is not the same as x , and we can pull out all the quantifiers to the front and have them have scope over the combined expression with the renamed variable.

Quantifier Equivalences



■ x where $F(x)$ is true
■ x where $G(x)$ is true

Equivalence formulas ($Q = \forall$ or $\exists$):

- $\underbrace{(\forall x F(x))} \wedge \underbrace{(\forall x G(x))} = \forall x \underbrace{(F(x) \wedge G(x))}$
- $\underbrace{(\exists x F(x))} \vee \underbrace{(\exists x G(x))} = \exists x \underbrace{(F(x) \vee G(x))}$

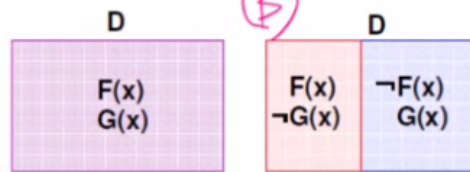
What about $(\forall x F(x)) \vee (\forall x G(x))$ and $\forall x (F(x) \vee G(x))$?

Let us look at these cases where we have the same variable shared between these two subexpressions. $\forall x F(x)$ and for all $x G(x)$. This would be satisfied by an example like that. For x in the domain, for every $x F(x)$ is satisfied, and for every $x G(x)$ is also satisfied.

Now let us look at this expression on the right-hand side, which says $\forall x F(x) \wedge G(x)$. So, this example that satisfies this expression will also satisfy this expression. So, they are equivalent.

Quantifier Equivalences

(A)



x where $F(x)$ is true
 x where $G(x)$ is true

Equivalence formulas ($Q = \forall$ or $\exists$):

- $(\forall x F(x)) \wedge (\forall x G(x)) = \forall x (F(x) \wedge G(x))$
- $(\exists x F(x)) \vee (\exists x G(x)) = \exists x (F(x) \vee G(x))$

What about $(\forall x F(x)) \vee (\forall x G(x))$ and $\forall x (F(x) \vee G(x))$?

(A): ok

(B): not ok

(A) ok

(B) ok

On the other hand, consider the case where this connective was instead of conjunction a disjunction, and we have these universal quantifiers with the same variable name. So, this says $\forall x F(x)$ or $\forall x G(x)$. So, this example here, A, will satisfy this expression, but this example here on the right will not satisfy this expression because if we pick an arbitrary x , either we want all of that to satisfy $F(x)$, but some of these would not satisfy $F(x)$.

And the same for $G(x)$. So, either of these two cannot be satisfied with this example.

Whereas this expression $\forall x F(x) \vee G(x)$, both examples A and B would satisfy this expression.

For this case, if we choose any arbitrary point x , it will satisfy $F(x)$. So, this will be satisfied.

Whereas if we select any arbitrary x , it would either satisfy $F(x)$ or $G(x)$. So, any point we choose within this domain will satisfy $F(x)$ or $G(x)$. So, this whole expression would be valid.

Quantifier Equivalences



□ x where $F(x)$ is true
□ x where $G(x)$ is true

Equivalence formulas ($Q = \forall$ or $\exists$):

- $\underbrace{(\forall x F(x))} \wedge \underbrace{(\forall x G(x))} = \forall x \underbrace{(F(x) \wedge G(x))}$
- $\underbrace{(\exists x F(x))} \vee \underbrace{(\exists x G(x))} = \exists x \underbrace{(F(x) \vee G(x))}$

What about $(\forall x F(x)) \vee (\forall x G(x))$ and $\forall x (F(x) \vee G(x))$?

We can think about similar examples like this, where these two are the same when we have existentially quantifiers with the same variable and a disjunction in the middle, but they are not the same when we have a conjunction in the middle.

Exercise: Conversion to Prenex Normal Form

Convert $(\forall x P(x)) \rightarrow (\exists y Q(y))$ to Prenex Normal Form:

$$\begin{aligned}
 (\forall x P(x)) \rightarrow (\exists y Q(y)) &= \neg(\forall x P(x)) \vee (\exists y Q(y)) \\
 &= (\exists x \neg P(x)) \vee (\exists y Q(y)) \\
 &= \exists x \exists y (\neg P(x) \vee Q(y))
 \end{aligned}$$

More exercise: $\forall x \forall y ((\exists z P(x, z) \wedge P(y, z)) \rightarrow (\exists u Q(x, y, u)))$

Let us try a small exercise. We want to convert $\forall x P(x) \leftrightarrow \exists y Q(y)$.

First, we have to get rid of the implication. So, we take the negation of that first part or the left-hand side of the implication and then connect that with the remainder with disjunction $\vee$. That is just a straightforward definition of implication. And then we use the trick. Moving this negation inside and flipping this quantifier.

So initially, it was universal, so we turned it into existential, and now we have $\exists x \neg P(x)$ with this scope $\vee \exists y Q(y)$ with this scope. And they have different variable names. So, we can just pull

these variables together to the front. So, we have $\exists x \exists y \neg P(x) \vee Q(y)$.

The scope of these two existential quantifiers would be this entire expression. We can also try this more challenging example and see if we can get it all to the front.

Example Proof: Motivating Skolemization

Given:

- 1 No used-car dealer buys a used car for his family.
 $\forall x (U(x) \rightarrow \neg B(x))$
- 2 Some people who buy used cars are absolutely dishonest.
 $\exists x (B(x) \wedge D(x))$
- 3 **Prove:** Some absolutely dishonest people are not used-car dealers.
 $\exists x (D(x) \wedge \neg U(x))$

Since the next step, which is the conjunctive normal form, is basically the same as the propositional logic case, let us move on to skolemization, which is the last step.

To motivate skolemization, we will look at this simple example theorem. No used-car dealer buys a used car for his family. Some people who buy used cars are absolutely dishonest. Based on these two premises, we want to prove: some absolutely dishonest people are not used-car dealers. Returning to first-order logic, this first sentence becomes $\forall x U(x) \leftrightarrow \neg B(x)$. U is the predicate for someone x being a used-car dealer.

And then $B(x)$ means someone x buys used cars for his family. The second sentence turns into $\exists x$ such that $B(x) \wedge D(x)$. Again, B means x buys used cars, and $D(x)$ means x is dishonest.

The last sentence we want to prove is $\exists x D(x) \wedge \neg U(x)$. Again, x is dishonest, and then x is not a used-car dealer.



Proof

$$\forall x (U(x) \rightarrow \neg B(x)) \quad (1)$$

$$\exists x (B(x) \wedge D(x)) \quad (2)$$

$$\text{Conclusion: } \exists x (D(x) \wedge \neg U(x)) \quad (3)$$

Assume (1) and (2) are true in domain D under interpretation I .

- ① Because of (2), there must be an x in D , which we shall call arbitrarily " a ", such that $B(a) \wedge D(a)$ is **True** under I .
- ② Thus, $B(a)$ is **True** and $\neg B(a)$ is **False**.
- ③ (1) is $\forall x (\neg U(x) \vee \neg B(x))$, so for $x = a$, $\neg U(a) \vee \neg B(a)$.
- ④ $\neg U(a)$ must be **True** because $\neg B(a)$ is **False**.
- ⑤ Because of (2), $D(a)$ is also true, thus $D(a) \wedge \neg U(a)$ is **True**, and " a " is one example where (3) is **True** in domain D .

Let us consider this theorem: $\forall x U(x) \leftrightarrow \neg B(x) \wedge \exists x$ such that $B(x) \wedge D(x)$. And we want to prove that $\exists x$ such that $D(x) \wedge \neg U(x)$.

So, let us assume these two are true. Because two is true, it means that within this domain, someone satisfies this condition $B(x) \wedge D(x)$. We do not know who that is, but because we assume this is true, we can think about that particular person who satisfies this condition and give that person a dummy name a .

From that, we can extract the single fact that $B(a) \wedge D(a)$ is true, which is again written here.

Because this is true, and for this to be true, both $B(a)$ and $D(a)$ must independently be **True** as well. From that, we know that $B(a)$ is **True**, and again, from that, we know that $\neg B(a)$ is **False**. So, it goes from $B(a)$ is **True** to $\neg B(a)$ is **False**.



Proof

$$\forall x (U(x) \rightarrow \neg B(x)) \quad (1)$$

$$\exists x (B(x) \wedge D(x)) \quad (2)$$

$$\text{Conclusion: } \exists x (D(x) \wedge \neg U(x)) \quad (3)$$

$$U(a) \rightarrow \neg B(a)$$

$$B(a) \wedge D(a)$$

$$B(a) : \text{True}$$

$$D(a)$$

$$\neg B(a) : \text{False}$$

Assume (1) and (2) are true in domain D under interpretation I .

- 1 Because of (2), there must be an x in D , which we shall call arbitrarily "a", such that $B(a) \wedge D(a)$ is **True** under I .
- 2 Thus, $B(a)$ is **True** and $\neg B(a)$ is **False**.
- 3 (1) is $\forall x (\neg U(x) \vee \neg B(x))$, so for $x = a$, $\neg U(a) \vee \neg B(a)$.
- 4 $\neg U(a)$ must be **True** because $\neg B(a)$ is **False**.
- 5 Because of (2), $D(a)$ is also true, thus $D(a) \wedge \neg U(a)$ is **True**, and "a" is one example where (3) is **True** in domain D .

Next, let us return to the first expression: $\forall x U(x) \leftrightarrow \neg B(x)$. This means that every x in this domain satisfies this condition, $U(x) \leftrightarrow \neg B(x)$. And then because a , the dummy person that we picked for the earlier example, also exists in the domain, from this, we can also see that $U(a) \leftrightarrow \neg B(a)$.

From that, just by removing this implication, we get $\neg U(a) \vee \neg B(a) = \text{True}$, but we already see that $\neg B(a) = \text{False}$. So, this would be $\neg U(a) \vee \text{False}$. So, for this to be true, $\neg U(a)$ has to be **True**. That is shown here.



Proof

$$\forall x (U(x) \rightarrow \neg B(x))$$

$$\exists x (B(x) \wedge D(x))$$

$$\text{Conclusion: } \exists x (D(x) \wedge \neg U(x))$$

$$U(a) \rightarrow \neg B(a)$$

$$\neg U(a) \vee \neg B(a) : \text{True}$$

$$\neg U(a) : \text{True (1)}$$

$$B(a) \wedge D(a) \quad (2)$$

$$B(a) : \text{True} \quad (3)$$

$$D(a) : \text{True}$$

$$\neg B(a) : \text{False}$$

Assume (1) and (2) are true in domain D under interpretation I .

- 1 Because of (2), there must be an x in D , which we shall call arbitrarily "a", such that $B(a) \wedge D(a)$ is **True** under I .
- 2 Thus, $B(a)$ is **True** and $\neg B(a)$ is **False**.
- 3 (1) is $\forall x (\neg U(x) \vee \neg B(x))$, so for $x = a$, $\neg U(a) \vee \neg B(a)$.
- 4 $\neg U(a)$ must be **True** because $\neg B(a)$ is **False**.
- 5 Because of (2), $D(a)$ is also true, thus $D(a) \wedge \neg U(a)$ is **True**, and "a" is one example where (3) is **True** in domain D .

With that, we have established that $\neg U(a) = \text{True}$, and from all here, we also have the fact that $D(a) = \text{True}$.

Combining these two, we have $D(a) \wedge \neg U(a)$ is also **True**. It turns out that if we apply that to this conclusion part, we see that this a , that is, when $x = a$, this expression $D(a) \wedge \neg U(a) = \text{True}$.

Thus, we deduced the fact that $\exists x$ such that $D(x) \wedge \neg U(x) = \text{True}$ given these two facts.

Proof

$$\forall x (U(x) \rightarrow \neg B(x)) \quad (1)$$

$$\exists x (B(x) \wedge D(x)) \quad (2)$$

$$\text{Conclusion: } \exists x (D(x) \wedge \neg U(x)) \quad (3)$$

Assume (1) and (2) are true in domain D under interpretation I .

- ① Because of (2), there must be an x in D , which we shall call arbitrarily " a ", such that $B(a) \wedge D(a)$ is **True** under I .
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- ③ (1) is $\forall x (\neg U(x) \vee \neg B(x))$, so for $x = a$, $\neg U(a) \vee \neg B(a)$.
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- ⑤ Because of (2), $D(a)$ is also true, thus $D(a) \wedge \neg U(a)$ is **True**, and " a " is one example where (3) is **True** in domain D .

This shows the usefulness of this kind of constant, some kind of anonymous constant that we use to make the deduction because we do not know that person or we cannot identify that person. But supposing that person exists, we can just give it a name, and we can follow through with this kind of inference. And this kind of dummy name or constant is called the skolem constant.

And this process, where we eliminate the variable or the existentially quantified variable and replace that with this dummy constant, is called skolemization.

Skolemization

Eliminate existential quantifiers through replacement of bound variables with **constants** or **functions**.

- Assume the formula is in Prenex Normal Form ($Q = \forall$ or $\exists$):
 $F = Q_1x_1 Q_2x_2 \dots Q_nx_n(M)$
- In a general case, where Q_rx_r is $\exists x_r$, delete Q_rx_r and replace every x_r in F by $f(x_{s_1}, x_{s_2}, \dots, x_{s_m})$, where $x_{s_1}, x_{s_2}, \dots, x_{s_m}$ are all the universally quantified variables appearing to the left of x_r , and f is a new function symbol (**Skolem function**) not appearing in F .
- If there is no universal quantifier $\forall$ to the left of the existential quantifier $\exists$ in question, remove $\exists$ and replace the variable associated with it with a constant a (called a **Skolem constant**).
- Observation: constant is like a function with no arguments.

Here is the full procedure for skolemization. We want to eliminate existential quantifiers through the replacement of bound variables with constants or functions. These could be skolem constants or skolem functions. Assume this formula F is in Prenex Normal Form, where Q could be either the universal quantifier or the existential quantifier.

So, the formula $F = Q_1x_1 Q_2x_2 \dots Q_nx_n$ is the matrix part. In the general case where we have Q_rx_r embedded in the middle, and when this quantifier Q_r is the existential quantifier, then delete Q_rx_r and replace every x_r in the matrix part with the function of $x_{s_1}, x_{s_2}, x_{s_m}$, where these variables are the universally quantified variables appearing in front of x_r .

So, for this function, F , we have to give it an arbitrary symbol name, F , G , et cetera. This is called the skolem function.

One specific case where we do not need to replace it with a function is when there are no universal classifiers to the left of the existential quantifier. And this was the case for the example we saw in the previous slide. So, in some sense, constants are like functions without any argument.

Skolemization Example

- **Initial formula:**

$$\exists x \forall y \forall z \exists u \forall v \exists w P(x, y, z, u, v, w)$$

- $\exists x \leftarrow a$, or equivalently, $h()$

- $\exists u \leftarrow f(y, z)$

- $\exists w \leftarrow g(y, z, v)$

- **Result of Skolemization:**

$$\forall y \forall z \forall v P(a, y, z, f(y, z), v, g(y, z, v))$$

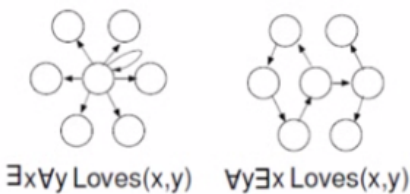
Let us consider this example written in Prenex Normal Form. And the matrix part is a single predicate, P : $\exists x \forall y \forall z \exists u \forall v \exists w$. And the predicate has x, y, z, u, v , and w as the arguments.

So, we go from the left to the right. The very first quantifier is an existential quantifier. So, we replace x with a . So, all occurrences of x in the following will be replaced by a . Then we move on past the universal quantifiers, and the next existential quantifier is associated with the variable u . The two universally quantified variables left of u are y and z , so we have to replace this u with a skolem function $f(y, z)$ where y and z are the two universally quantified variables appearing to the left.

And we go past another universal quantifier. And finally, we reach here. We have an existentially quantified variable w . Again, we find that there are three universally quantified variables, y, z , and v . So, to replace w , we have to construct another anonymous function. In this case, let us call it $g()$ to avoid any conflict with $f()$ from earlier. So, we replace w with $g(y, z, v)$.

So finally, what we get is this skolemized expression where these quantifiers in the front are all universally quantified variables. So, P of a, a was the skolem constant and y, z are universally quantified variables. And $f(y, z)$ is the skolem function, v is the universally quantified variable, and $g(y, z, v)$ is another skolem function.

Quantifier Order and Skolemization



Different quantifier order results in different Skolemization:

- $\exists x \forall y \text{ Loves}(x, y)$
 $= \forall y \text{ Loves}(a, y)$
- $\forall y \exists x \text{ Loves}(x, y)$
 $= \forall y \text{ Loves}(f(y), y)$

a is a new Skolem constant, and $f(\cdot)$ is a new Skolem function.

This may seem a bit abstract, so let us look at this skolemization process in the context of these two examples we saw earlier. Earlier, we saw that we cannot change the order of these universal and existential quantifiers like that because the meaning will now differ. There exists x such that for all y , this particular person x loves y .

So, here it is shown that this particular person x would satisfy this expression and would love everyone else, including himself or herself. Whereas when we switch the order of these two quantifiers, then it acquires a totally different meaning. So, $\forall y \exists x$ such that someone x who exists loves this y that is for everyone.

So, if we pick anyone, y , this person is loved by someone else. If we select this y , then this person is loved by someone, and so on. We can show that every one of these people is loved by someone. So, these two are very different.

So, even without thinking about these different examples, just by simply skolemizing these two expressions, we can easily see what the difference is. So, in this case, $\exists x \forall y \text{ Loves}(x, y)$. Because this appears in the front without any universally quantified variables ahead of it, we can replace this with a skolem constant, a .

So, this turns into: $\forall y \text{ Loves}(a, y)$, and then this person shown here would be that example, a . Whereas when we skolemize the second expression: $\forall y \exists x \text{ Loves}(x, y)$. Then once we reach here, there is another universally quantified variable, y , to the left. So, instead of replacing this x with a constant, we must replace it with a function. To be specific, a function of y .

So, this expression becomes $\forall y \text{ Loves}(f(y), y)$. So, for every y here, there is someone who loves y , which is a function of this particular choice, y , and that is exactly represented here.

Standard Form: A Summary

We followed these three steps to convert first-order logic formulas into a standard form amenable to algorithmic verification:

- 1 Transform formula into **Prenex Normal Form**.
- 2 Transform the matrix into **Conjunctive Normal Form**.
- 3 Eliminate existential quantifiers through **Skolemization**.

⇒ A set of **clauses** in **CNF** in which all variables are **universally quantified**

Here is a brief summary. In this lecture, we follow these three steps. Prenex normal form, Conjunctive normal form, and Skolemization to convert first-order logic formulas into standard form, amenable to algorithmic verification. That is, once we have turned these expressions into the standard form, it is very easy for the machine algorithm to deal with them.

As a result, what we get is just a set of clauses in Conjunctive normal form where all variables are universally quantified. So again, we are back to the collection of clauses in the knowledge base.

Standard Form: A Summary

We followed these three steps to convert first-order logic formulas into a standard form amenable to algorithmic verification:

- 1 Transform formula into **Prenex Normal Form**.
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- 3 Eliminate existential quantifiers through **Skolemization**.

⇒ A set of **clauses** in **CNF** in which all variables are **universally quantified**



So, it will be a little bit different, but here we have a clause with two predicates with universally quantified variable x . Because every variable is universally quantified, we can simply drop the universal quantifier symbol. And the second clause, we have $\neg R(x, y)$, again, x and y are universally quantified, and so on.

So, we have just a bunch of clauses. And as we can expect, we can resolve this by picking two clauses with one complementary literal, reducing that, and adding it back.

We will look at a very small example proof in the next lecture.

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MODULE 9

STANDARD FORM (CANONICAL FORM)